ZIYANG CHEN

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EDUCATION

University of Michigan, Ann Arbor Jan. 2022 - 2025 (Expected)
Ph.D. candidate in Electrical & Computer Engineering | Advisor: *Andrew Owens*

University of Michigan, Ann Arbor Sept. 2019 - May 2021
Master of Science in Engineering, Computer Vision
Overall GPA: **4.0/4.0** | Advisor: *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sept. 2015 - Aug. 2019
Bachelor of Science in Engineering
Major: Electrical & Computer Engineering | Minor: Entrepreneurship
GPA: **3.60/4.0** with honors & *Graduated magna cum laude*

PUBLICATIONS

- *Ziyang Chen, Daniel Geng and Andrew Owens.* "Images that Sound: Composing Images and Sounds on a Single Canvas". (NeurIPS 2024) (Project site)
- *Chao Feng, Ziyang Chen, Aleksander Holynski, Alexei A Efros, and Andrew Owens.* "GPS-to-3D: Lifting Tourist Photos to 3D Using 2D GPS-Conditioned Diffusion". (In submission)
- *Ziyang Chen, Israel D. Gebru, Christian Richardt, Anurag Kumar, William Laney, Andrew Owens, and Alexander Richard.* "Real Acoustic Fields: An Audio-Visual Room Acoustics Dataset and Benchmark". (CVPR 2024, **Highlight**) (Project site)
- *Fengyu Yang*, Chao Feng*, Ziyang Chen*, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong.* "Binding Touch to Everything: Learning Unified Multimodal Tactile Representations". (CVPR 2024) (Project site)
- *Ziyang Chen, Shengyi Qian and Andrew Owens.* "Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation". (ICCV 2023) (Project site)
- *Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.* "Conditional Generation of Audio from Video via Foley Analogies". (CVPR 2023)(Project site)
- *Chao Feng, Ziyang Chen and Andrew Owens.* "Self-Supervised Video Forensics by Audio-Visual Anomaly Detection". (CVPR 2023, **Highlight**) (Project site)
- *Ziyang Chen, David F. Fouhey and Andrew Owens.* "Sound Localization by Self-Supervised Time Delay Estimation". (ECCV 2022) (Project site)
- *Xixi Hu*, Ziyang Chen* and Andrew Owens.* "Mix and Localize: Localizing Sound Sources in Mixtures". (CVPR 2022) (Project site)
- *Ziyang Chen*, Xixi Hu* and Andrew Owens.* "Structure from Silence: Learning Scene Structure from Ambient Sound". (CoRL 2021, **Oral**) (Project site)

WORK EXPERIENCE

Research Scientist Intern, *Adobe Research, San Francisco* May. 2024 -
- Supervisor: Prem Seetharaman, Justin Salamon

Research Scientist Intern, *Meta Reality Lab, Pittsburgh* May. 2023 - Nov. 2023
- Topics: neural acoustic fields
- Supervisor: Israel D. Gebru, Christian Richardt, Alexander Richard

Research Assistant, *University of Michigan, Ann Arbor* Jan. 2020 - Present
- Topics: multi-modal learning, self-supervised learning

Teaching Assistant, *Shanghai Jiao Tong University* Sept. 2018 - Aug. 2019

- Courses: *Digital Integrated Circuits*, *Electronic Circuits*
- Conducted experiments, held office hours and taught recitation classes every week

Research Engineer Intern, *FreeSense Image Inc., Shanghai* Mar. 2018 - May 2018

- Implemented an auxiliary system for a project of Detection of Car LCD Screen and applied it into industrial production lines
- Adopted state-of-the-art recognition DNNs to improve the accuracy of defect detection tasks

SERVICE

Reviewer

- Conference: WACV 2023, CVPR 2023, ICCV 2023, CVPR 2024, SIGGRAPH 2024, ECCV 2024, NeurIPS 2024
- Journal: IJCV

RESEARCH EXPERIENCE

Images that Sound Jan. 2024 - Sep. 2024
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Propose *images that sound*, a type of multimodal art that can be both understood as an image or played as a sound
- Demonstrate the zero-shot composition of pretrained diffusion models across modalities can generate examples at the intersection of image and spectrogram distributions. *(NeurIPS 2024)*

Real Acoustic Fields May. 2023 - Feb. 2024
Research scientist intern, Codec Avatars Lab, Meta *Mentor: Israel D. Gebru, Alexander Richard*

- Collect a dataset called Real Acoustic Fields (RAF) that captures real acoustic room data from multiple modalities
- Conduct experiments to benchmark existing acoustic field methods *(CVPR 2024)*

Binding Touch to Everything Aug. 2023 - Feb. 2024
Project Lead, University of Michigan *Advisor: Andrew Owens*

- Proposed a method can learn unified multimodal tactile representations for multiple sensors by learning from images
- Applied this representation to various zero-shot and cross-modal applications without any further training *(CVPR 2024)*

Sound Localization from Motion Aug. 2022 - May 2023
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a self-supervised method to jointly learn to estimate sound source direction and camera pose
- Proposed an audio-visual representation learning through a self-supervised cross-view binauralization task *(ICCV 2023)*

Video Forensics by Audio-Visual Anomaly Detection Jan. 2021 - Mar. 2023
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a zero-shot audio-visual anomaly detection method by leveraging the synchronization between audio and frames in the real videos *(CVPR 2023)*

Conditional Foley Generation June 2021 - Mar. 2023
Research Associate, University of Michigan *Advisor: Bryan Russell, Justin Salamon, Andrew Owens*

- Proposed a self-supervised conditional Foley model that creates audio tracks for silent videos conditioning on given videos and sounds *(CVPR 2023)*

Sound Localization by Self-Supervised Time Delay Estimation June 2021 - June 2022
Research Associate, University of Michigan *Advisor: David F. Fouhey, Andrew Owens*

- Proposed self-supervised methods to learn interaural time difference from Internet stereo or mono audio
- Proposed a novel visually-guided time delay estimation task: localizing a speaker in a multi-speaker recording
- Evaluated on simulated environments with metrically accurate ground truth and on internet videos with directional judgments achieving significantly better results *(ECCV 2022)*

Mix and Localize July 2020 - March 2022
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a simple self-supervised audio-visual method using contrastive random walk with cycle-consistency constraints and applied learned representations to sound source localization task
- Implemented state-of-the-art sound localization methods as baselines
- Evaluated our method on human speech and musical instruments datasets and achieved significantly better qualitative and quantitative results *(CVPR 2022)*

Structure from Silence Jan. 2020 - May 2020, Apr. 2021 - Sep. 2021
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Collected datasets with ambient sound and paired RGB-D videos of the indoor environment
- Demonstrated that the ambient sound could infer geometric properties with experiments and proposed a new audio-visual learning method to learn to exploit structural representation
- Apply learned models to robots and navigate them via ambient sound only in the real world *(CoRL 2021)*

AWARDS & SCHOLARSHIPS

- **Lums Fellowship**, *University of Michigan, Ann Arbor* 2019-2021
- **Outstanding Graduate - Class of 2019**, *Shanghai Jiao Tong University* 2019
- **Academic Excellence Scholarship**, *Shanghai Jiao Tong University* 2016-2018
- **Dean's List**, *Shanghai Jiao Tong University, SJTU-UM JI* 2015-2018